

principle of parsimony to estimate phylogenies in a similar way: They examine large numbers of possible trees and identify those that require the fewest evolutionary changes.

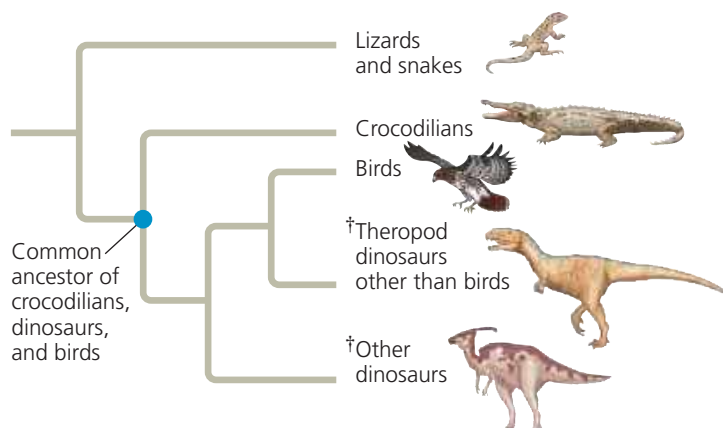
Phylogenetic Trees as Hypotheses

This is a good place to reiterate that any phylogenetic tree represents a hypothesis about how the organisms in the tree are related to one another. The best hypothesis is the one that best fits all the available data. A phylogenetic hypothesis may be modified when new evidence compels systematists to revise their trees. Indeed, while many older phylogenetic hypotheses have been supported by new morphological and molecular data, others have been changed or rejected.

Thinking of phylogenies as hypotheses also allows us to use them in a powerful way: We can make and test predictions based on the assumption that a particular phylogeny—our hypothesis—is correct. For example, in an approach known as *phylogenetic bracketing*, we can predict (by parsimony) that features shared by two groups of closely related organisms are present in their common ancestor and all of its descendants unless independent data indicate otherwise. (Note that “prediction” can refer to unknown past events as well as to evolutionary changes yet to occur.)

This approach has been used to make novel predictions about dinosaurs. For example, there is evidence that birds descended from the theropods, a group of bipedal dinosaurs. As seen in **Figure 26.16**, the closest living relatives of birds are crocodiles. Birds and crocodiles share numerous features: They have four-chambered hearts, they “sing” to defend territories and attract mates (although a crocodile’s “song” is more like a bellow), and they build nests. Both birds and crocodiles also care for their eggs by warming them. Birds warm their eggs by sitting on them, whereas crocodiles cover

▼ **Figure 26.16 A phylogenetic tree of birds and their close relatives.** (A dagger, †, indicates extinct lineages.)



VISUAL SKILLS In this tree diagram, what is the sister taxon of the clade that includes dinosaurs and their most recent common ancestor? Explain.

their eggs with their neck. Reasoning that any feature shared by birds and crocodiles is likely to have been present in their common ancestor (denoted by the blue dot in **Figure 26.16**) and *all* of its descendants, biologists predicted that dinosaurs had four-chambered hearts, sang, built nests, and cared for their eggs.

Internal organs, such as the heart, rarely fossilize, and it is, of course, difficult to test whether dinosaurs sang to defend territories and attract mates. However, fossil discoveries have supported the prediction that dinosaurs built nests and cared for their eggs. First, a fossil embryo of an *Oviraptor* dinosaur was found, still inside its egg. This egg was identical to those found in another fossil, one that showed an *Oviraptor* crouching over a group of eggs in a posture similar to how birds sit on their nests today (**Figure 26.17**). Researchers suggested that the *Oviraptor* dinosaur preserved in this second fossil died while incubating or protecting its eggs. The broader conclusion that emerged from this work—that dinosaurs built nests and cared for their eggs—has since been strengthened by additional fossil discoveries that show that other species

▼ **Figure 26.17 Fossil support for a phylogenetic prediction: Dinosaurs built nests and cared for their eggs.**

